

Siyan Wang

Email: ldb24001@uconn.edu

Storr, CT

EDUCATION

University of Connecticut (UConn)

Ph.D student of Statistics

Storr, CT

09/2024–current

- Interest: Extreme value, Risk control
- Skills: Statistical computing, Applied Statistics, Optimization

University of Wisconsin–Madison (UW–Madison)

Master of Statistics and Data Science (MSDS)

Madison, WI

09/2022–05/2024

- Skills: Statistical Learning, Deep Learning, Statistical Visualization

Huazhong Agricultural University (HZAU)

Bachelor of Science in Information and Computation Science

Wuhan, China

09/2018–06/2022

- Skills: Mathematical Analysis, Machine Learning, Data Mining

RESEARCH EXPERIENCE

Research Assistant, University of Connecticut

Student Success Analysis

Office of Vice president for Student Life and Enrollment

09/2024–current

- Developed predictive modeling pipelines to identify students at elevated risk of DFW outcomes by using selected and created variables.
- Constructed multi-level academic indicators and scenario-based evaluation workflows to examine risk heterogeneity across students, programs, and course contexts.
- Designed visualization and reporting workflows to support institutional decision-making on student success and course-level intervention strategies.

Research Assistant, HZAU

Study on the Characteristics of Medical Soft Tissue Based on Fractal Dimension

College of Science

10/2019–05/2022

- Led a five-member research team on quantitative CT image analysis for lung disease differentiation, including COVID-19 and lung cancer.
- Designed a fractal-dimension-based feature extraction pipeline for 10,000+ CT images across whole-lung and lung-parenchyma regions.
- Benchmarked supervised and unsupervised learning models, with SVM achieving the strongest diagnostic performance at high accuracy.

Research Assistant, HZAU

Study on Drought Stress of Fruit and Vegetable Seedlings

College of Engineering

12/2020–09/2021

- Investigated whether plants lacked water based on outcomes including leaf color, shape, and height.
- Collected depth images and RGB images from different angles using a Kinect tool and infrared camera.
- Calculated the FD of each image and studied the relationship between water content and FD.

PUBLICATION

- Jingfang Shen, Sijie Cheng, **Siyan Wang**, and Wenwei Liu. “A Multi-Scale Stiffness Fractal Model of Joint Interfaces.” *Russian Physical Journal*, 2019.

RESEARCH PROJECTS

Body Fat Prediction Analysis in Different Age Groups

10/2023

Course Project, UW–Madison

- Built predictive models for estimating body fat using diverse physiological features and analyzed the distribution of body fat across different age groups.
- Revamped a Shiny app to visually showcase prediction results and create an engaging interactive website for prospective users.
- Contributed to the concept and execution of the age-group analysis and Shiny app construction.

Fruit Classification Based on Various Models and Activation Functions

05/2023

Course Project, UW–Madison

- Implemented multilayer perceptron (MLP) and convolutional neural network (CNN) models for classification on a dataset of 10 fruit classes.
- Built a CNN model by adapting AlexNet to fit fruit image classification.
- Fine-tuned CNN parameters and optimized hyperparameters, including kernel sizes, pooling layers, and activation functions.
- Built and modified models and compared their classification efficiency.

Application of Multidimensional Scaling Algorithm (MDS)

09/2020–01/2021

Course Paper, HZAU

- Verified the feasibility of MDS in data visualization by displaying the relative positions of points on a two-dimensional coordinate plane based on the distance matrix between U.S. cities.
- Compared the MDS visualization results with the cities’ actual locations on a map.
- Explored two-dimensional images of outlier data after MDS dimensionality reduction and concluded that MDS visualization becomes less reliable under partial outliers.

AWARDS & PRIZES

- National Scholarship, HZAU (Top 1%), 2019.
- Excellent Student, HZAU (Top 10%), 2019, 2020, 2021, and 2022.

TRAINING

University of Oxford, Exeter College

08/2019

Global Leadership Programme

- Proposed a company acquisition programme and won first place in the final project presentation.
- Studied International Economics, Game Theory, and Globalization.